Introduction to Web & Mobile App Development

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Introduction

- Goal: to learn those **basic tools** for the *design and implementation* of web applications using HTTP for the client/server communication.
- Main topics to work on:
  - Classical web applications
  - Interactive applications
  - Mobile applications

Main topics

- Web applications:
  - Servlets
  - Persistence
  - MVC pattern
- Interactive applications:
  - Polling/pushing options
  - Publisher/subscriber pattern
  - RESTful web services and WebSockets
- Mobile applications:
  - User interface
  - Communication with a remote server
  - Local persistance

Laboratory sessions

- Required software:
  - Java SE (JDK8) (version 8!!! otherwise NetBeans 8.2 doesn't work)
  - NetBeans IDE 8.2, J2EE edition
  - MySQL 5.7 Community Server (version 5.7!!! otherwise the ddbb connection from NetBeans 8.2 doesn't work (install only the server!!!)
  - MySQL Workbench, any version
  - Android Studio IDE 2.3.3. and emulator-API 21/22/23 (otherwise websocket library problem)

Required setup

- Due to a bug of GlassFish 4.1.1. you have to go to the directory where you have the GlassFish 4.1.1 installed and access to the subfolder: "glassfish-4.1.1\glassfish\modules" to:
  - add the file: jackson-module-jaxb-annotations-2.5.1.jar
  - replace the file: org.eclipse.persistence.moxy.jar
- Define a new GlassFish server and corresponding domain, for the changes to take effect, check the default checkbox at the ports windows to use 8080.

Required setup

- Import the database ecommerce.sql
- Create the user/password: ecomm/ecomm, and give it DBA privileges.
- Define the ecommerce database connection from NetBeans.

Required setup

- Create a web project.
- Create an entity package.
- Create new entity classes from database.
- Create a web application listener using:

```
@Resource
public UserTransaction utx;
@PersistenceContext
public EntityManager em;
```

- Run the web project as it is only to check it is deployed properly, nothing else.

Required setup

- After installing the Android Studio IDE 2.3.3: for the Android exercise you have to:
  - "Import project (Eclipse ADT, Gradle, etc.)"

  do not try to start/open/import code sample.
- After creating the exercise project, update it as requested.
- Create an Emulator only with API 21/22/23.
- Run the exercise as it is onto the emulator.

Subject assessment

- Submission of the following exercises:
  - Virtual grocery
  - Message wall service (2 versions)
  - Instant messaging service (3 versions)
  - Chat mobile application
- Individual interview at the end of the term if necessary

Exercises weights

- 20% eCommerce
- 10% messageWall
- 35% instantMessaging (15% / 10% / 10%)
- 35% whatsApp